COURSEWORK ASSIGNMENT SUBMISSION

Name: Daksh Srivastava **Roll No:** 150096725087
Subject: Software Engineering & Project Management **Day:** Day 20 — Software Quality Assurance, Metrics & Maintenance

---

Section 10: Classroom Activity — Quality Dashboard Interpretation

Analysis of staging release metrics for CEMS Release 1.2 to formulate a release recommendation.

Quality Metric	Reported Measurement	Engineering Evaluation & Action
Executed Test Cases	90 Passed, 10 Failed (90% pass rate)	The 10 failures must be triaged. If any failure affects core registration, release must be blocked regardless of the 90% overall pass rate.
Open Defects	2 Critical, 5 Major, 8 Minor	CRITICAL RELEASE BLOCKER. Zero critical defects are tolerated in production. The 2 critical bugs (data loss or crash) must be resolved and re-tested.
Performance Latency	p95 Response Time = 2.4 seconds	Fails agreed SLA target (target is < 1.5 seconds). Indicates un-optimized database joins or lack of query caching under load.
System Availability	99.0% over observation window	Equivalent to 14.4 minutes of downtime per day. Substandard for active registration week; requires root-cause analysis of web server crashes.
Post-Release Tickets	6 maintenance requests	Classified as 2 corrective defects, 3 UI adaptations, and 1 feature enhancement request.

Process Improvement Recommendations: (1) Institute an automated Continuous Integration (CI) regression gate that automatically halts build deployments if any critical test fails; (2) Implement database query indexing and Redis response caching to bring p95 latency below the 1.5-second threshold.

Section 13: Homework — Quality Plan & Maintenance Classification

Quality assurance governance and maintenance taxonomy for campus enterprise software.

1. Four-Tier Software Quality Assurance Plan

- **Quality Goal 1 (Defect Density):** Maintain defect density below 0.5 defects per KLOC prior to production deployment, verified via static code analysis (SonarQube).
- **Quality Goal 2 (Automated Test Coverage):** Enforce minimum 80% line and branch code coverage on business logic services as a mandatory pull-request merge gate.
- **Quality Goal 3 (Performance Benchmarking):** Conduct automated weekly load tests verifying that p95 latency remains under 1.5 seconds at 300 concurrent simulated users.
- **Quality Goal 4 (Security Compliance):** Execute bi-weekly OWASP Top 10 vulnerability scans with zero unresolved high or critical security alerts.

2. Test Pass Rate Analysis

Reported: 84 passing tests out of 100 executed tests (84% pass rate). Mathematical calculation: $(84 / 100) \times 100 = 84\%$. Operational Limitation: The pass rate percentage fails to reveal defect severity. If the 16 failing tests all reside in trivial CSS layout formatting, the software may be operationally viable. However, if even 1 of the 16 failing tests represents an unhandled database transaction crash during seat reservation, the system is completely unfit for release.

3. Maintenance Taxonomy Classification

Maintenance Request Description	Maintenance Category	Engineering Rationale
Fix crash when user submits special characters in name field.	Corrective Maintenance	Removes an active coding bug causing system malfunction.
Update authentication provider to support college Google Workspace OIDC.	Adaptive Maintenance	Modifies system to function in an altered external environment.
Redesign event discovery card to display speaker photographs and bios.	Perfective Maintenance	Adds enhanced functional value and improves user experience.
Refactor database queries and introduce indexing before festival surge.	Preventive Maintenance	Hardens system against anticipated future performance failures.
Patch OpenSSL security vulnerability in container base image.	Corrective / Preventive	Eliminates known security vulnerability before exploitation.